AYON ROY

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Summary

Ph.D. candidate in Computer Science with over 3 years of research experience in machine learning, and multi-modal data analysis, with 2+ years in full stack software development. Expertise in designing advanced deep learning models for cognitive assessment, human-computer interaction, and agentic AI applications. Published researcher with awards for innovation and academic contributions, passionate about advancing Generative AI and intelligent systems to address real-world challenges.

Education

The University of Texas at Arlington *Aug 2023 – Present*
Ph.D. in Computer Science
Military Institute of Science and Technology (MIST) *Jan 2017 – Dec 2020*
BS in Computer Science

Technical Skills

Languages: Python, C, C++, Java, GO, JavaScript, Typescript, SQL

Libraries/Frameworks: NextJS, ReactJS, NodeJS, Material UI, Redux, TailwindCSS, Angular, Flutter, Django, Flask

Databases: PostgreSQL, Firebase, MongoDB, DynamoDB

ML/AI: PyTorch, TensorFlow, Keras, Numpy, Pandas, Matplotlib, Scikit-learn, Ollama, n8n, opal

Tools: Git, Docker, Kubernetes, Google Colab, Visual Studio, Android Studio

Experience

The University of Texas at Arlington *Arlington, TX*
Graduate Teaching Assistant & Research Assistant *Aug 2023 – Present*

- Self-supervised representation learning from multi-modal vision data (images, EEG, EDA, LiDAR)
- Developed a Residual Graph Convolutional Network (ResGCN) to detect cognitive fatigue level from gait cycle data that reflects walking patterns
- GTA for intermediate C, JAVA, and OOP courses

Trinity Industries *Dallas, TX*
Data Analytics Intern *May 2026 – July 2026*

- Built and deployed an end-to-end LLM + ML system that automated multi-output repair-cost classification, pairing Claude Sonnet clause extraction over 1,292 OCR-parsed lease contracts with a CatBoost ensemble trained on 4M+ records, deployed to a Databricks/MLflow serving endpoint that scored 62,715 line items and replaced manual analyst review
- Built and optimized a 4-model ensemble forecasting railcar out-dates on Databricks, engineering 60+ features and combining KMeans segmentation (automated k via knee detection), CatBoost classifiers, and per-cluster random forests that return prediction intervals, routing each car to its best-fit sub-model for daily confidence-bounded forecasts used by shop operations

Techzu *Singapore*
Full Stack Developer *Oct 2021 – July 2023*

- Built data-driven solutions with React and React Native apps along with MongoDB, helping users access cross-platform responsive features
- Developed full-stack projects using Django and PostgreSQL, making it easier to manage and store user data securely

NetraDigital *Toronto, Canada*
Machine Learning Engineer (Part-Time) *Sep 2021 – Feb 2022*

- Developed ML and RL based models to determine risk factors in projects, which helped the team catch potential issues early and plan better
- Built an automated workload management pipeline, making task assignments faster and helping the team stay on top of priorities

Selected Publications

- ChatGPT versus Bard: A comparative study. *Engineering Reports, vol. 6, no. 11, e12890*, 2024
- Examining the landscape of cognitive fatigue detection: A comprehensive survey. *Technologies, vol. 12, no. 3, p. 38*, 2024
- CurFi: An automated tool to find the best regression analysis model using curve fitting. *Engineering Reports, vol. 4, no. 12*, 2022
- A Deep Learning Approach to Predict Academic Result and Recommend Study Plan for Improving Student's Academic Performance. *Ubiquitous Intelligent Systems*, 2021

HONORS & AWARDS

- **Dean's Research Award in 2024**, UTA CSE Department
- **Doctoral Consortium Award**, PETRA 2025 and PETRA 2024, Greece
- **First Prize winner** for the poster presentation on UTA Innovation Day 2024